ASSIGNMENT

Introduction to Symmetric Cryptography

Subject: Cryptography and Network Security

Q1. Explain the basic principles of symmetric cryptography and Kerckhoffs's principle.

Symmetric cryptography uses the same secret key for encryption and decryption. The sender converts plaintext into ciphertext using an encryption algorithm and the secret key. The receiver uses the same key with the decryption algorithm to recover the original plaintext. The key must be shared securely.

Plaintext is the original readable message; ciphertext is the encrypted message; the secret key controls the transformation; encryption converts plaintext to ciphertext; and decryption restores plaintext.

Block diagram:
Plaintext → [Encryption algorithm + Secret key] → Ciphertext → [Decryption algorithm + Same secret key] → Plaintext

Kerckhoffs's principle says a cryptographic system should remain secure even if its design and algorithm are public. Security should depend on keeping the key secret, not on hiding the algorithm.

Q2. Distinguish stream ciphers and block ciphers. Explain confusion, diffusion and ECB, CBC, CFB, OFB and CTR modes.

A stream cipher encrypts data one bit or byte at a time, usually by combining it with a keystream. A block cipher processes fixed-size blocks of data. Stream ciphers suit continuous data, while block ciphers are widely used for files and network protocols.

Confusion makes the relationship between the key and ciphertext complex, commonly through substitution. Diffusion spreads the influence of each plaintext bit across many ciphertext bits through permutation and mixing.

• ECB: Encrypts each block independently. It is simple and parallelizable, but identical plaintext blocks create identical ciphertext blocks and reveal patterns.
• CBC: XORs each plaintext block with the previous ciphertext block before encryption; the first block uses an IV. It hides repeated patterns better than ECB, but encryption is sequential and padding may be needed.
• CFB: Uses cipher output as feedback to encrypt data in segments. It can work like a stream cipher and needs no padding, but errors may affect following segments.
• OFB: Generates a keystream from repeated cipher operations. Bit errors do not propagate through later plaintext, but the IV must not be reused with the same key.
• CTR: Encrypts successive counter values to create a keystream. It is fast and parallelizable and needs no padding, but counter/nonce reuse with the same key is dangerous.

Modern applications often prefer authenticated modes such as GCM because they can provide confidentiality and integrity together.

Q3. Trace the historical development of symmetric encryption algorithms.

Early methods included the Caesar cipher and other substitution/transposition ciphers. During World War II, the Enigma rotor machine protected communications; its cryptanalysis involved mathematics, machines, captured information and operational weaknesses. Claude Shannon's work on information theory and secrecy systems laid important foundations for modern cryptography and introduced concepts such as confusion and diffusion. The Feistel structure used repeated rounds operating on halves of a block and influenced DES.

DES was adopted as a U.S. standard in 1977. Its effective 56-bit key eventually became too short against brute-force attacks. Triple DES applied DES multiple times to improve security, but was slower. Other algorithms included IDEA and Blowfish (block ciphers) and RC4 (a stream cipher now considered insecure). An open competition selected Rijndael, which became the Advanced Encryption Standard (AES) in 2001.

Q4. Explain Caesar, monoalphabetic substitution, Playfair and Vigenère ciphers. Why is frequency analysis effective against some?

(a) Caesar cipher: Each letter is shifted by a fixed number. With a shift of 3, CAT becomes FDW. It is easy to use but has few possible keys and is easy to break by trying them all.

(b) Monoalphabetic substitution: Each plaintext letter is consistently replaced by another letter. Its key space is large, but it preserves letter-frequency patterns, so frequency analysis can break it.

(c) Playfair cipher: Encrypts pairs of letters using a 5×5 key square. Letters in the same row are replaced by letters to their right; letters in the same column by those below; otherwise the rectangle rule is used. It hides single-letter frequencies better, but digraph patterns can still leak information.

(d) Vigenère cipher: Uses a repeating keyword to apply different Caesar shifts to successive letters. It reduces simple single-letter frequency patterns, but short repeating keys can be identified using methods such as Kasiski examination, followed by frequency analysis.

Frequency analysis works well when a cipher preserves language statistics, as Caesar and monoalphabetic substitution do. It is less direct against polyalphabetic ciphers, though repeating-key patterns may still reveal information.

Q5. Describe rail fence and columnar transposition, product ciphers and the one-time pad.

A transposition cipher rearranges character positions without changing the characters. In Rail Fence, the message is written diagonally across several rails and read row by row. In columnar transposition, plaintext is written in rows under a keyword and columns are read in the order determined by that keyword. Both preserve letter frequencies and can be attacked using language patterns and trial arrangements.

A product cipher combines substitution and transposition. Substitution changes symbol values, while transposition changes positions; repeated rounds provide confusion and diffusion and form the basis of many modern block ciphers.

A one-time pad combines plaintext with a truly random key of equal length, used only once. If the key is secret, random and never reused, it provides perfect secrecy in theory. It is impractical because long random keys must be securely generated, distributed and stored. Reusing the key can expose information.

Q6. Describe DES structure, its Feistel structure, sizes, permutations, rounds, round function and key schedule.

DES is a symmetric block cipher with a 64-bit block size and a supplied 64-bit key, of which 8 bits are parity bits; the effective key length is 56 bits. It uses 16 Feistel rounds. The initial permutation rearranges the input bits, which are split into two 32-bit halves, L0 and R0.

In each round, the right half is expanded from 32 to 48 bits, XORed with a 48-bit round key, passed through eight S-boxes that reduce it to 32 bits, and rearranged by a P-box permutation. The result is XORed with the left half. The update is Li = Ri−1 and Ri = Li−1 XOR F(Ri−1, Ki). After 16 rounds and the final half arrangement, the inverse initial permutation produces ciphertext.

Key schedule: the effective 56 bits are permuted and split into two 28-bit halves. The halves are shifted left by specified amounts each round, and a compression permutation selects 48 bits to form each round key.

DES round diagram:
R(i−1) [32 bits] → Expansion E [48 bits] → XOR with Ki [48 bits] → S-boxes [32 bits] → P permutation [32 bits] → XOR with L(i−1) → R(i)
At the same time, L(i) = R(i−1). Decryption uses the same Feistel structure with round keys in reverse order.

Q7. Discuss DES strengths and weaknesses, avalanche effect, brute force, cryptanalysis, Double DES and Triple DES.

The avalanche effect means that changing one input or key bit should change many output bits, typically about half after sufficient rounds. DES has this useful property, but its 56-bit effective key is too short against modern brute-force search. Differential cryptanalysis studies how plaintext differences affect ciphertext differences, while linear cryptanalysis uses statistical relationships among plaintext, ciphertext and key bits.

Double DES encrypts twice: C = E(K2, E(K1, P)). Although it appears to provide a 112-bit key space, a meet-in-the-middle attack matches intermediate values computed using possible K1 and K2 keys, reducing its security advantage.

Triple DES applies DES three times, commonly in encrypt-decrypt-encrypt form. Two-key 3DES uses K1, K2, K1; three-key 3DES uses independent K1, K2 and K3. It improves security over single DES but is slower and retains a 64-bit block size. Modern systems generally prefer AES.

Q8. Explain AES, key lengths, rounds, transformations, key expansion, decryption and comparison with DES.

AES is a symmetric block cipher based on Rijndael. It has a fixed 128-bit block size and supports keys of 128, 192 and 256 bits, using 10, 12 and 14 rounds respectively. Data is arranged as a 4×4 byte state.

- SubBytes: Replaces each byte using a nonlinear S-box.
- ShiftRows: Cyclically shifts rows by different offsets.
- MixColumns: Mixes bytes in each column using finite-field arithmetic; it is omitted in the final encryption round.
- AddRoundKey: XORs the state with a round key derived from the original key.

Encryption begins with AddRoundKey, followed by rounds of SubBytes, ShiftRows, MixColumns and AddRoundKey; the final round omits MixColumns. Key expansion derives a round key for each stage. Decryption uses inverse transformations (InvShiftRows, InvSubBytes, AddRoundKey and InvMixColumns) in the appropriate order with round keys applied in reverse order.

DES uses 64-bit blocks, a 56-bit effective key and a 16-round Feistel structure. AES uses 128-bit blocks, keys of 128/192/256 bits and a substitution-permutation network. AES provides stronger modern security and is generally more efficient on current hardware; DES is no longer secure because its key is too short.

References

1. William Stallings, Cryptography and Network Security: Principles and Practice.

2. NIST, FIPS PUB 46-3, Data Encryption Standard (DES).

3. NIST, FIPS PUB 197, Advanced Encryption Standard (AES).

4. Claude E. Shannon, Communication Theory of Secrecy Systems, 1949.